Efficient Electric Vehicles Assignment for Platoon-based Charging

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***Abstract*—To fulfil the increasing charging requests of Electric Vehicles (EVs), various means have been proposed such as improving charging efficiency at charging stations, optimal locating of charging stations, enabling V2V trading and so on. Most existing work requires vehicles to be stationary while being charged. Mobile wireless charging is a promising trend to fix this problem since vehicles can perform moving and charging simultaneously. In this paper, a new concept called platoon-based charging is presented, which combines the energy-aware driving and mobile charging. In the design, an EV may have to detour to follow the platoon. To minimize the additional energy cost and delay brought by that, we studied the optimal assignment problem between EVs and charging platoons and converted it to a dynamic weight bipartite matching. The experiment results show that our algorithm outperforms the exiting ones.**

I. Introduction

With the rapid increasing of various vehicles on road, the carbon emissions and fossil fuel consumption are estimated to be doubled within the next 10 to 20 years, which puts heavy burdens on environment protection and resource preservation of the earth. Electric vehicles have been proved to be eco-friendly and experienced a fast growth in many countries [1]. Currently, there are still some factors that discourage drivers from choosing EVs. The first reason is that EVs have to face the inconvenience of recharging: significant amount of charging time compared with fuel based vehicles which can be refilled within a few minutes; limited choices of charging station while petrol stations are everywhere. The second reason is that EVs running on electricity last less miles than traditional vehicles because of the capacity of lithium-ion batteries. Thus, to improve the recharging efficiency is one of the most important issues on which many researchers have been working.

Numerous solutions have been proposed to solve this problem, from charging station locationing [2][3] to charging sequence scheduling [4], from cost-aware V2G charging [5] to V2V energy trading [6], from eco-routing of EVs [7] to mobile wireless charging [8], and so on. Apart from satisfying their own demanding, vehicles can even act as transporters of energy which forms so called Vehicular Energy Network [9]. With respect to the transportation efficiency, it is more feasible to enable energy exchange while vehicles are moving. One way is to build new infrastructures, such as in ref. [10], two online electric vehicle buses have been set up in the city of Gumi by Korea Advanced Institute of Science and Technology. The experimental road segments are embedded with magnetic fields so that vehicles can get charged while driving on them. In China, a one-kilometer-long high way was built in 2017, which further utilizes photovoltaic power generation to provide energy to EVs running on it. However, the cost of building such kind of roads is very high and it also needs tremendous time to update current roads. The other way is to employ a mobile charging vehicle, such as a bus or truck [8]. These mobile energy disseminators (MEDs) are running along major road segments with fixed or flexible routes. Other EVs can find these MEDs and book charging places trough wireless communications. This technology is more practical according to the current situation but there are still a lot to improve. For example, the EVs will find a MED in its own interests which may result local optimal since EVs will compete for MEDs. Also, each EV acts as an isolated unit while charging, this fails to take the charging position into consideration and charging capacity is not maximized for each MED.

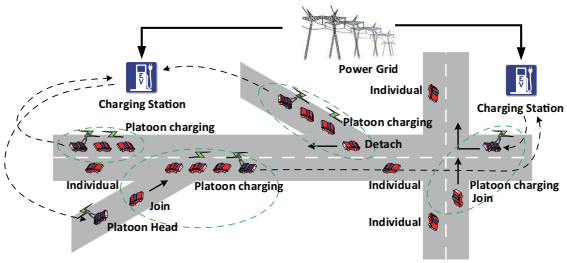

Fig. 1. A platoon charging demo.

In recent years, the formation of a group of vehicles at close inter-vehicular distances, known as a platoon, has attracted a lot of research activities since it can reduce energy consumption, increase road safety, and decrease traffic congestion [11]. Therefore, platoon based driving pattern facilitates the potential of platoon based charging where the platoon head also acts as the MED. In this way, both road and energy efficiency can be achieved. As shown in Fig. 1, there are several charging platoons running in the area. Platoon heads are equipped with batteries of large capacity. They first go to charging stations to replenish their energy and then driving along the road waiting for EVs to join in. The individual EV will submit its destination, initial power level and delay deadline to the sever.

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The latter one will assign the EV to the appropriate platoon and help it to join and detach at designated location. The route planning of platoon heads could be based on many metrics such as the demanding and trajectory of EVs. However, this is beyond the scope of this paper. We assume that the optimal routes of platoon heads have already been given. The focus of the paper is how to optimally assign each EV to a charging platoon so as to maximize the global performance. There remains some challenges, e.g., the different position in a platoon results in different charging bandwidth according to the wireless charging property, the balance between the acquired energy and the detour. In this paper, we propose a novel charging platoon assignment method. Our contribution is threefold:

1) We propose to use platoon to charge EVs so that both road and energy efficiency are achieved.
2) We consider the different energy charging (receiving) rates in terms of the position in the platoon.
3) We extend the maximum-weight assignment solution for the dynamic matching cost problem.

Summarizing, this work is structured as follows: in Section 2, we present the model and the general formulation of the dynamic weight multiple assignment problem. Next, in Section 3, the details of our method implementation are given. After that, in Section 4, we validate our results through synthetic simulations. Finally, we conclude our paper with a summary and future work in Section 5.

II. Related work

In many existing work, EVs are charged at home [12], at charging stations or through V2V energy trading [13][6]. In [14], a balanced charger station assignment is proposed. Ref. [8] is a similar work compared with ours but it adopts a FCFS strategy. Compared with traditional fossil fuel based vehicles which can be fully refueled within several minutes, EVs need a significant amount of time for charging, as a result of limited charging facilities and battery technology [15]. Once an EV is short of energy on road, no matter it seeks a nearby charging station or a parking lot for V2V energy trading, considerable delay and detour cost can be expected. Therefore, besides current V2X and V2V charging technologies [16], we need to develop a more efficient way to enable EV charging in a more flexible and cost-efficient manner.

Wireless power transfer is a promising technology which enables non-contact charging. It does not require the physical connection through a power cord between the charger and the receiver EV. However, it still suffer from low efficiency, thus requires the charger and the receiver EV be close to each other. With the advance of new technologies, one can envision that the performance of wireless charging will have a significant improvement in the near future. In the vehicular platoon, vehicles maintain a small and nearly constant distance to the preceding vehicle, because a platoon-based driving pattern can significantly improve road capacity and energy efficiency [11]. Quite a lot previous work has addressed research aspects such as platoon modeling, platoon management, platoon stability control, cooperative platoon driving, platoon-based V2V/V2I communication, etc. Platoon-based driving has been put in applications such as highway grouped driving (SARTRE program [17]), green traffic (Energy ITS [18]), and data delivery (V-PADA [19]). In the future, platoon-based driven will be more popular since it incorporates the autonomous vehicles [20][21].

III. System Model

A. System overview

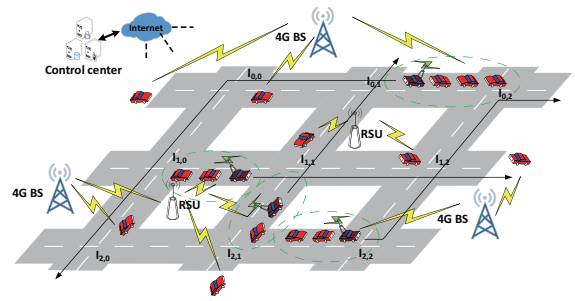

Fig. 2. System architecture of the platoon based charging

We consider a platoon-based EV charging scenario as shown in Fig. 2, which is a typical application of the emerging technology: Internet of Energy [22]. Energy plants, stationary charging stations, and power grids are omitted from the proposed scenario. Although they provide the backbone support, we mainly focus on optimizing of platoons assignment for EVs. In the system, the planar road segments could be derived from downtown area of any city. Each intersection of roads is numbered (e.g., I_{01}) so that we can easily describe the trajectory of vehicles. Road side units and 4G Network are common in nowadays vehicular networks so that EVs, platoon heads, charging stations, and the control center can easily communicate with each other and exchange information about energy price, electricity demands, trajectory and location in a real-time manner [23]. We do not adopt the multi-hop vehicle to vehicle communication model since the message delay is not tolerable in this scenario. There are four major entities and terms in the proposed system, as shown below.

Receiver EV, R-EV, an EV in need of energy, denoted by $RV_i, i = 1, 2, ..., N$. It will post its energy demand and trajectory to the control center. An assignment will be returned to schedule it to a vehicle platoon for charging.

Platoon head, a special vehicle, denoted by $H_j, j = 1, 2, ..., M$, is equipped with a large capacity battery sets, a wireless or wired charging device, and wireless communication components. It moves in the area along the fixed or demand-responding trajectory and charges R-EVs following it.

Charging platoon, in addition to a general vehicle platoon which purpose is driving efficiency and safety on highways [17], the charging platoon is composed of an electricity-ample platoon head and several R-EVs. An EV joins a platoon when being scheduled by the control center, and detaches

the platoon upon finishing charging. The charging platoon is denoted by $P_k, k = 1, 2, ..., K$.

Control center, it collects demands from R-EVs and their trajectories, applies the optimal algorithm to match R-EVs with charging platoons, so that R-EVs' energy demands can be fulfilled with minimal cost and delay.

B. Wireless charging model

There are typically four types of wireless charging technologies as shown in table. I. According to current progress of technology, most of them only support a short charging range such as a few center meters. At the meantime, the charging power is less than 10w which is far beyond the requirement of EV charging. The Resonance is the only means that fits for the proposed scenario. It supports several kW charging power and the distance of a few meters. However, the charging efficiency decreases sharply with the increasing of distance. Therefore, EVs which are not directly following the MED will hardly get any energy. In this paper, we propose multi-hop wireless energy charging model, as shown in Fig. 3. The EV in the middle of the platoon will receive the energy from foregoing neighbour and transmit the rest energy to its later neighbour. Due to transmission loss and relays' own needs, the energy can be transmitted will decrease from hop by hop. Suppose the wireless energy transmission loss rate is ζ, and each EV will keep $BW(RV_i)$ portion of energy. Then, the i-th EV can get $\leq (1-\zeta)^i - (1-\zeta)^{i-1}BW(RV_i) - (1-\zeta)^{i-2}BW(RV_2) - ... - (1-\zeta)BW(RV_{i-1})$ energy. It depends on the energy transmission efficiency and the EV's place in the charging platoon. With the advance of technologies, the model will become more practical. For example, provided that in the future, the wireless charging technology can support larger power and further distance, the model still works as the energy which i-th EV can get only relies on distance. In another word, it is just with different parameter. Nevertheless the case, we use charging bandwidth to describe the energy that the platoon head can provide as well as the energy EVs can get in terms of the distance (hops).

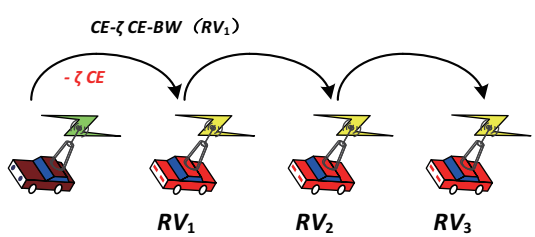

Fig. 3. A multi-hop charging model.

C. Platoon-based charging

The vehicle platoon itself is in favour of mobile electric charging. First, the vehicle platoon is moving directly or indirectly to the destination of the R-EV so that the latter one does not necessarily waste its time. Second, vehicles in the platoon are close to each other so that the R-EV can utilize both wired or wireless charging means. The last but not the least, the platoon remains relatively stable, as a result, the R-EV can have a fully usable charging time. To sum up, the platoon-based charging sounds reasonable to be implemented.

We will not go through the detail of the platoon-based driving since it is out of the scope of this paper. Instead, this paper is going to tackle some challenges before efficiently putting platoon based charging into practice. For the platoon head, the charging bandwidth should be considered, i.e., the total power subjects to the physical constraints where the R-EVs must share the energy transfer capability. We model the charging bandwidth as the rate of energy charging that a R-EV can be assigned either due to multi-hop or distance effect. The charging bandwidth itself could be fixed or floating. Regarding the former one, each R-EV gets fixed and guaranteed charging resources. Regarding the latter one, the more the R-EVs, the less the charging resources can each of them get.

Furthermore, for the R-EVs, the charging detour should also be considered. Some EVs may have to change their path to follow the charging platoon so that there will be extra energy consumption and delay. The best deal is the trajectory of the platoon is highly aligned with that of the R-EV while it can provide enough charging bandwidth. However, that can not always be true since there are limited resources v.s. various EV preferences. To achieve the global of charging efficiency, each R-EV should be assigned to a proper charging platoon. Without loss of generality, we assume that the battery capacity of the platoon head is always enough since it may be equipped with large volume batteries and can be charged at charging stations during off-duty. In the rest of this paper, we will propose a platoon-based charging management method to achieve optimal charging efficiency for the entire system.

D. Problem formulation

In the proposed scenario, the energy demand of EVs is evolving along the time scale. Energy requests can be posted anytime and anywhere via the stat-of-the-art wireless communications. After collecting these requests, the control center will assign these EVs to the proper charging platoons according to their trajectories and charging bandwidth to make the best of their energy gain against lost due to the charging detour. The detailed problem formulation will be described below.

1) timing of assignment: Apparently, there is no one-time assignment policy, as the energy requests will be posted along the time scale. From another point of view, the assign-as-it-request is also of big overhead. Therefore, we divide time into time slots. Only one assignment will be applied in each time slot. The length of a time slot depends on two facts, i.e., the more requests we have in a time slot the better assignment can be achieved for them, on the other hand, the shorter the time slot, a R-EV can get quicker response. According to Fig. 2, road planar is described as a grid. EVs will decide their directions at each road intersection. Hence, the best length of a time slot is defined as the time it needed for driving through a road segment between two intersections. In the proposed

TABLE I
FOUR TYPICAL WIRELESS CHARGING TECHNOLOGIES

Technology	Magnetic induction	Resonance	Radio reception	Capacitive coupling
Transmission power	$\leq$ 5W	a few KWs	100mW	1-10W
Frequency	22KHz	13.56MHz	2.45GHz	560-700KHz
Efficiency	80%	50%	38%	70%-80%
Distance	mm to cm	cm to m	$\geq$ 10m	mm to cm
Features	Short dist, high efficiency	Long distance, large power, moderate efficiency	Long distance, low power, easy to use	Short distance, high efficiency
Provider	Ti, Powermat, Splashpower	MIT, Intel, Fujitsu	Powercast	Murata

algorithm and the simulation, we use the median of different length of road segments.

2) assignment based resource allocation: In this paper, we tend to adopt the means of fixed charging bandwidth. Once a R-EV RV_i is assigned to a charging platoon P_j, the platoon head H_j will set aside fixed charging bandwidth for the EV so that the latter one can get guaranteed service. Notice that for each R-EV RV_i, it has its maximal energy receiving rate, denoted by $maxC_i$. The allocated charging bandwidth $BW(RV_i, P_j)$ (Def. 1) can not be greater than that. For simplicity, we regard that all R-EVs have the same maximal energy receiving rate. If the remainder charging capacity is not enough to reach $maxC_i$, e.g., multiple EVs compete for the residual available charging bandwidth, there are two methods to tackle this problem. The first one is to always assign $maxC_i$ to newly joined EV RV_i. Once the residual charging capacity is smaller than $maxC_i$, the platoon will reject the request of the R-EV RV_i. The strategy is simple but not optimal. For example, in the worst case, the R-EV RV_i may be rejected by all charging platoons while their residual charging capacities are wasted. The other method is that the multiple EVs equally share the remaining charging capacity, which is adopted in this paper. In this case, every portion of energy is utilized and a R-EV will always find a platoon to get charged. If there are l R-EVs going to be assigned to the charging platoon P_j, and RV_i is any one of those l R-EVs, we have:

Definition 1: The charging bandwidth for R-EV RV_i at the charging platoon P_j, denoted by $BW(RV_i, P_j)$, is defined as a guaranteed energy transfer rate between the platoon head and the R-EV during the charging period. The measuring unit is Joule per Minute, i.e., J/M.

$$BW(RV_i, P_j) = \begin{cases} maxC_i, & if\ \frac{CE_j^t}{l} \geq maxC_i \\ \frac{CE_j^t}{l}, & otherwise \end{cases} \quad (1)$$

where CE_j^t is defined as Def. 2.

Definition 2: The charging capacity of the platoon head H_j at time slot t, denoted by $CE_j^t, t = t_0, t_1, ..., t_T$, is defined as the remainder charging bandwidth that can be allocated for the R-EVs. The measuring unit is Joule per Minute, i.e., J/M. The initial capacity without being allocated by any EVs is denoted as $CE_j^{t_0}$.

3) resource release of assignment: After a R-EV finishes its charging procedure, it will release the charging bandwidth that has been allocated to it. Sometimes, there is only a short period of time before the R-EV reaches its destination. Therefore, adequately charged or arriving at the destination, whoever is first, the receiver will quit the charging platoon and release the resource. In other cases, the route of the charging platoon may be quite different with that of the R-EV. It will evaluate the energy gain and cost, and decide to quit at the right time.

Except above conditions, once the resource is allocated to the R-EV, the latter one will keep holding it. In other words, we do not reallocate the entire resources in every time slot. At the beginning of each time slot, only remaining available resource will be assigned to newly joined EVs. There are three major reasons. First one is the computing cost. It will incur a considerable computing overhead if the algorithm is executed too frequently. The second one is that joining in and detaching from the charging platoon needs some time, which is not negligible compared with the length of the time slot. The third one is that reallocation cannot achieve the optimal outcomes. To summarize, the current optimal assignment is based on the current condition assuming it will not change. If some assignments change afterwards, it breaks the foundation of previous assignment. When an EV misses the charging platoon, it cannot get back since they may drive to different directions. The only optimal solution is to utilize the history which requires accurate prediction of the future. However, the future is difficult to predict because the estimation based on probability is not suitable for precise matching.

In most occasions, the trajectory of the charging platoon is not as exactly the same as the driving route of the R-EV towards its destination. More or less, the R-EV will need to detour as to follow the charging platoon. We have the definition below.

Definition 3: The charging detour for R-EV RV_i following the charging platoon P_j, denoted by $DE(RV_i, P_j)_t^{t+k}$, is defined as the additional journey the R-EV has to travel (including detour before, during, and post the charging period), when it is charged by the charging platoon from time t to $t+k$ for k slots of time. The measuring unit is meter.

IV. DYNAMIC-WEIGHT EV AND PLATOON MATCHING ALGORITHM

According to the Platoon-based charging model, in this section, we propose a bipartite matching based algorithm to maximize the system performance. The goal is to minimize charging cost and latency from the view point of the whole system. The classical Kuhn-Munkres theorem and Hungarian algorithm [24] is targeting at max-weight and one-to-one matching. To solve the proposed problem, we extend the

classical algorithm to support many-to-one assignment and dynamic weight constraints.

Before we go to the algorithm, here we prepare some preliminary knowledge such as definitions, notations and parameters. First, we need to identify the benefit and cost of the R-EVs if they are assigned to the potential charging platoon. The benefit is the charging bandwidth that is allocated to the EV. The larger the bandwidth, the quicker the EV can get charged. As stated above, once the bandwidth is allocated, it will not change until the EV is detached from the charging platoon. The cost is the charging detour of the EV, which is a discrete function of time. We use $DE(RV_i, P_j)^t$ to represent the charging detour at time t. Now we have the total charged energy of an assign between a RV_i and P_j pair for $t_{RV_i}^{P_j}$ time slots

$$E_c(RV_i, P_j) = t_{RV_i}^{P_j} BW(RV_i, P_j) - DE(RV_i, P_j)_t^{t+t_{RV_i}^{P_j}} \alpha_i \quad (2)$$

where α_i is the moving energy consumption per meter for RV_i. Since $k_{RV_i}^{P_j}$ is a variable to different charging platoons, the second thing is to decide appropriate $t_{RV_i}^{P_j}$ for each RV_i and P_j pair. The time delay for this assignment will be (assume that all EVs have the same speed v):

$$Delay(RV_i, P_j) = \frac{DE(RV_i, P_j)_t^{t+t_{RV_i}^{P_j}}}{v} \quad (3)$$

Then the total utility of assigning RV_i to P_j for $t_{RV_i}^{P_j}$ time slots is:

$$U(RV_i, P_j) = \frac{E_c}{Delay} = t_{RV_i}^{P_j} BW(RV_i, P_j) v - \alpha_i v \quad (4)$$

Finally, we want to maximize the total utilities of all R-EVs. Since α_i and v are constant values, we have:

$$\max \sum_{i=1}^{n} U(RV_i, P_j) = \max \sum_{i=1}^{n} \frac{t_{RV_i}^{P_j} BW(RV_i, P_j)}{DE(RV_i, P_j)_t^{t+t_{RV_i}^{P_j}}} \quad (5)$$

Depending on which and how many R-EVs are assigned to a platoon P_j, the parameter $t_{RV_i}^{P_j}$, $BW(RV_i, P_j)$ and $DE(RV_i, P_j)_t^{t+t_{RV_i}^{P_j}}$ change accordingly which is a typical dynamic weight assignment and it can be solved by applying the algorithm in our previous work [25].

V. Simulation

To evaluate the performance of the proposed model and algorithm, we compare the proposed algorithm with the one in ref.[8] through simulation based experiments. The simulation scenario is based on Fig. 2. A 3km $\times$ 3km area is setup and totaly 100 EVs are put into it. The initial energy of EVs follows the Gaussian distribution with parameters u and α (In the evaluation, u is set to 0.4 and α is set to 0.5, which means many of the EVs are hungry). The arrival rate of the EVs follows the Poison process with parameter λ (In the simulation, λ is set to 6). Upon entering this area, each EV will randomly choose an Entrance and an Exit from the 12×12 matrix.

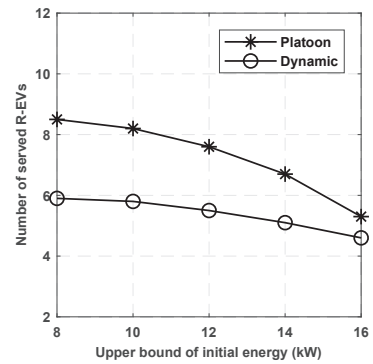

Fig. 4. The performance of different initial energy, served R-EVs.

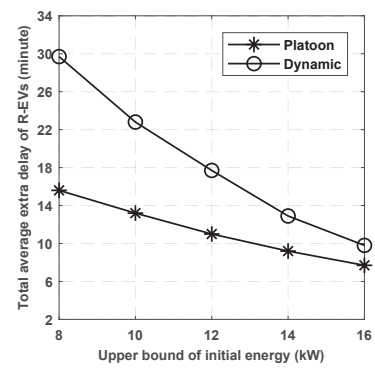

Fig. 5. The performance of different initial energy, average extra delay.

We mainly concentrate on two metrics, the first one is the total served R-EVs, the second one is the extra delay due to detour. Fig. 4 shows the relationship between the initial energy and the number of served R-EVs. When the upper bound of initial energy is low, there are more R-EVs which are in need of energy. When the upper bound of initial energy is 8kW, our algorithm can support around 40% more R-EVs. With the increasing of the initial energy, the number of severed R-EVs will decrease. It is mainly because there are less demanding from R-EVs. We show the extra delay induced due to mobile charing in Fig. 5. When the initial energy is 8kW, our algorithm can save about 48% of time and always outperform the competitor. The reason is that the waiting time for charging platoon in ref. [8] greatly worsen the delay.

We also evaluate the performance under different charging capacity of MEDs. Since the adopted technology and equipment may be different, each MED can support limited number of R-EVs. Fig. 6 shows that when there are more charging requests than the system capability, an increase in the energy capacity leads to more successfully served R-EVs. Our platoon-based algorithm achieves 35% better performance than

dynamic-based algorithm in terms of number of served R-EVs. Fig. 7 tells that the average extra delay has little to do with the energy capacity when there are too many starving R-EVs. From the experiment, we draw a conclusion that our algorithm outperforms the one based on dynamic charging with a overall performance increasing around 40%.

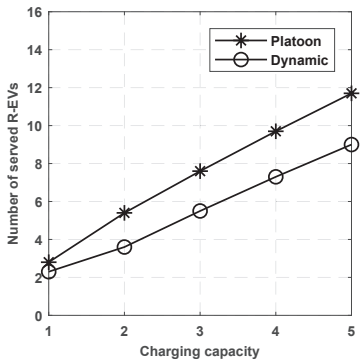

Fig. 6. The performance of different energy capacity, served R-EVs.

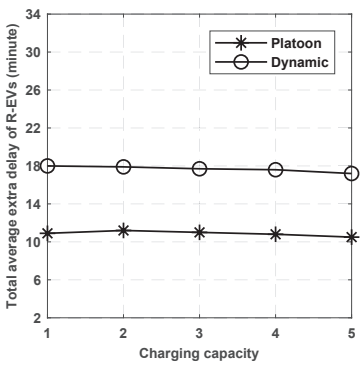

Fig. 7. The performance of different energy capacity, average extra delay.

VI. Conclusion

In this paper, we have proposed a novel platoon-based charging strategy to facilitate EVs to extend their driving range. We have analyzed the challenges of implementing this strategy. State-of-the-art communication technologies are applied to enable information exchange among the control center, platoon heads and EVs such as energy request, trajectory, speed and charging capacity. Based on those, an efficient EV assignment method is designed. Simulation results demonstrate that the proposed method can achieve much better performance.

Acknowledgment

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